GraphIV retrieval audit — every step, every constant

inputs, algorithm, reasoning, evidence status, and what (if anything) to sweep

2026-07-19

One section per pipeline step, in execution order. Each gray box lists every constant the step uses, its current value, and where the value came from: **measured** (an experiment picked it), **inherited** (taken from literature with evidence there, unverified on our data), or **hand-set** (a guess that has never been swept). Status tags: **BACKED** = evidence exists (internal experiment and/or literature), **PARTIAL** = the mechanism is backed but its constants are not, **UNBACKED** = no evidence either way. "Cheap" sweeps are deterministic pool-stage runs on the silver-qrels pilots (no LLM, minutes per condition); "standard" sweeps need graded runs (GPU, ∼15–25 min each).

Literature receipts come from the 2026-07-19 GraphIV run `retrieval-strategies-for-scientific-lit-66358b01` (pool 258, 26 grounded citations) plus classic IR references.

Terminology

term	meaning
KG	the Postgres store: `paper` + intent-typed `citation` + embeddings + topic tree
V	vector pool — pgvector KNN over asym embeddings, size `n_pool`
L	lexical pool — Postgres full-text over titles, size `k_lexical`
RRF	reciprocal rank fusion, $1/(60 + \text{rank})$ summed over lists
m	landmark budget: papers selected into the citable cohort (60/80/120 by tier)
landmark	selected paper, citable in the report
connector	pool paper kept only as a graph edge endpoint, never cited
rel	reader relevance grade $-2..+2$; fusion score = cos $\times(\text{rel} + 2)/4$
role	reader 6-class label: foundational / core / frontier / adjacent / survey / unrelated
PPR	personalized PageRank on the induced subgraph ($\alpha = 0.85$) — baseline selector
Leyline sampler	magnetic Monte-Carlo BFS selector ($q = 8$, $\gamma = 0.4$, 10 seeds); formerly "trajectory"
leyline	one lineage thread traced by a walker; convergence across seeds = selection
magnet	rel ≥ 2 paper pulling walkers, weight $(\text{rel} + 2)/4$
frontier year	per-topic recent-activity cutoff from the pool's own year histogram
recency r	0..1 knob reserving round$(r \cdot m)$ slots for frontier-year papers
ancestor floor	min cosine 0.30 for backward-citation augmentation (coupled to $m \geq 60$)
tier	effort preset bundling `n_pool`/m/extract: tiny / fast / standard / deep

1 Step 0 — query embedding

Inputs: the question string.

constant	value	provenance
prompt prefix	`search_query:`	nomic model contract
embedding space	asym (`paper_embedding_asym`)	**measured** (2026-07-15 A/B vs sym+HyDE)
HyDE	off in asym mode	**measured** (same A/B)
endpoint	local llama-embed, mean pooling	parity vs ST cosine > 0.999

Algorithm. One HTTP call to the local embedding server; the vector is the search key for every later step.
Why. Asym prompts already encode the query/document role split, so HyDE's "rewrite the question as a fake abstract" trick is redundant.
Backed? **BACKED** — internal A/B chose asym; endpoint parity verified.

Action: none.

2 Step 1 — candidate pools $V \cup L$, fused by RRF

Inputs: qvec, question string.

constant	value	provenance
$\lvert V\rvert$ = `n_pool`	200 / 400 / 400 / 600 (tiny/fast/std/deep)	hand-set tiers
$\lvert L\rvert$ = `k_lexical`	400 (all tiers)	hand-set
lexical field	`tsvector` over *titles only*	hand-set
`RRF_K`	60	inherited (Cormack et al. 2009)
output size	top `n_pool` of fused rank	follows $\lvert V\rvert$
`hnsw.ef_search`	1000	**measured** (02-07; default 40 truncated every KNN)

Algorithm. V = pgvector KNN top-`n_pool`. L = Postgres full-text rank over titles. Fused score per paper:

$$s(p) = \frac{1}{60 + \text{rank}_V(p)} + \frac{1}{60 + \text{rank}_L(p)},$$

keep the top `n_pool`. L never grows the pool — it only reshuffles membership.
Why. Cosine and BM25 scores live on incomparable scales, so fuse on ranks. The +60 damps single-list dominance: rank 1 in one list (1/61) loses to rank 30 in both (2/90), i.e. agreement between two weak signals beats one signal's confidence. At k=0 one channel's favorite steamrolls consensus; at $k \to \infty$ the formula degenerates to "in how many lists does it appear."
Backed? **PARTIAL**. RRF itself: *Cormack, Clarke & Büttcher, SIGIR 2009* (tuned k=60 once on TREC, found insensitive; industry default since). Hybrid dense+sparse for scientific docs: *arXiv:2401.04055*, *arXiv:2508.01405*. Our hybrid arm was promoted from kg_rag_lab as the polysemy fix. But `n_pool`, `k_lexical`, k=60, and titles-only were never swept on our data.

Action: cheap sweeps on the 3 pilot surveys: `n_pool` $\in \{200, 400, 600, 800\}$, `k_lexical` $\in \{100, 400, 800\}$, `RRF_K` $\in \{10, 60, 300\}$ against core-ref pool recall. One extra probe: titles-only is the most suspicious choice here — try title+abstract `tsvector`, since survey refs often match on abstract vocabulary.

3 Step 2 — topic-adaptive frontier detection

Inputs: year histogram of the fused pool, current date (or the `--set-max-date` cap), landmark budget m, recency knob.

constant	value	provenance
`_WINDOW`	7 years	hand-set
`_ACTIVE_SHARE`	0.30 of recent peak	hand-set
`_MAX_SPAN`	6 years	hand-set
`recent_floor`	round(recency $\cdot\, m$)	formula; recency knob **measured** (exp-2)
recency default	0.4	**measured** (exp-2 sweep)

Algorithm. Find the peak year-count within the last 7 years; walk back from now while each year holds $\geq$30% of that peak (the part-elapsed current year gets a pass); clamp to at most 6 years back. Papers

≥ that year are "frontier"; `recent_floor` landmark slots are reserved for them, filled newest-cohort-first round-robin (`frontier_select`) so 2025 *and* 2026 each get slots.
Why. A fixed cutoff (the old hardcoded 2023) buried new work in hot fields and starved slow fields; each topic's own tempo should define "recent." Newest-first round-robin exists because similarity-ranked reservation systematically fed the just-past-cutoff years and starved the true frontier.
Backed? **PARTIAL**. The recency *knob* is measured (exp-2); the three detector constants are not. Literature: the 07-19 corpus run found *nothing* on recency-adaptive retrieval despite it being in the question verbatim — this is one of the three claimable novelties, which raises the evidence bar.

Action: do not sweep 7/0.30/6 blindly. The battery (20 capped surveys, different topic tempos) will exercise the detector; log `frontier_year` per topic and eyeball for wrong calls. Sweep only if one looks wrong. The paper needs at least a sensitivity sentence.

4 Step 3 — leaf anchor + citation drill + recency inject

Inputs: fused pool, topic tree, qvec, frontier year.

constant	value	provenance
leaf-vote seeds `topk`	30	hand-set
`use_siblings`	true (keep leaf ∪ same parent path)	hand-set
drill `target`	150 / 280 / 280 / 360 by tier	hand-set tiers
`recent_k` (inject)	50	hand-set
`member_cap`	5000 leaf members read	hand-set

Algorithm. (a) similarity-weighted vote among the top-30 pool papers picks the dominant leaf and parent path; (b) *anchor*: drop pool papers outside leaf-or-siblings; (c) *drill*: refill toward `target` with leaf members that cite / are cited by the surviving pool, ranked by cosine; (d) *inject*: add the top-50 leaf members ≥ frontier year, newest-cohort-first.
Why. The anchor kills polysemous-query hijack (spintronics → cond-mat, not particle physics — the validated case that motivated it). The drill restores topical depth the anchor removed. The inject counters the tree's old-paper bias.
Backed? **PARTIAL**, and this is the step in genuine doubt. The channel as a whole is backed by exp-1 (`no_cluster` ablation) and is family-adjacent to cluster-based GraphRAG *arXiv:2404.16130*. BUT study 04-03 found `no_cluster` *improved* pool recall on the diffusion-SR survey (0.775 vs 0.518 at ≥2 mentions) — on a tight, non-polysemous question the anchor appears to *evict true references*. The hard anchor itself has no literature antecedent (novelty candidate *and* risk).

Action: highest-priority retrieval question for the battery: run a `no_cluster` arm across all 20 surveys. Expected outcome: anchor helps polysemous/broad questions, hurts tight ones — which argues for a *conditional* anchor (fire only when the top-30 leaf vote is split) rather than always-on. Measure before changing the design.

5 Step 4 — frontier pull (leaf-independent)

Inputs: qvec, frontier year.

constant	value	provenance
`probe`	1500 nearest neighbors scanned	hand-set
`want`	max(40, `recent_floor`)	hand-set
selection	newest-cohort-first round-robin	shared `frontier_select`

Algorithm. KNN top-1500 over the *whole* index, keep papers ≥ frontier year, take `want` of them newest-first.
Why. The topic tree predates the corpus delta, so the newest papers have no leaf and step 3 can never reach them. This is the only channel that can admit them.
Backed? **BACKED** — exp-3, and 04-03 re-confirmed on silver qrels (removing it dropped R@1 0.518 → 0.473).

Action: one diagnostic only: in the battery, log how many of the 1500 probed survive the year filter. If a hot topic's recent output saturates the probe, the pull sees a biased slice and `probe` needs raising. No sweep otherwise.

6 Step 5 — backward-ancestor augmentation

Inputs: pool, qvec.

constant	value	provenance
`max_add`	120	hand-set
`min_citers`	3 distinct pool citers	hand-set
`min_cosine`	0.30	**measured** (exp-16 / 02-03 → 02-07 BYOL)
coupling	floor 0.30 requires $m \geq 60$	**measured** (02-07; never ship apart)

Algorithm. Rank out-of-pool papers by number of distinct pool papers citing them (fame *ranks*); admit up to 120 that pass the 0.30 cosine floor (relevance *gates*).
Why. The foundational lineage (DDPM, score-SDE, . . .) is cited by nearly the whole pool yet is rarely a top KNN neighbor of the question — pure similarity retrieval structurally misses origins. The floor keeps famous cross-field hubs out.
Backed? **BACKED** — the most thoroughly measured channel: floor swept (0.35 Pareto on a 0.25–0.35 plateau), lowered to 0.30 on BYOL evidence with the documented m coupling. Literature: backward snowballing is standard systematic-review methodology (*Wohlin 2014*; *arXiv:2307.02612* shows search+snowballing beats either alone); seed-set citation ranking ≈ co-citation analysis (*Small 1973*).

Action: none. `min_citers`=3 and `max_add`=120 are unswept but the channel sits on a measured plateau; not worth compute.

7 Step 6 — subgraph induction

Inputs: final pool ids.

constant	value	provenance
edge collapse	strongest intent wins (`INTENT_PRIORITY`)	hand-set ordering
self-loops	dropped	—

Algorithm. One SQL fetch of all citation rows with both endpoints in the pool; parallel edges collapse to one carrying the strongest-signal intent + its polarity.
Why. Everything downstream (PPR, magnets, story graph, digest adjacency) consumes this typed mini-graph.
Backed? **BACKED** — exp-1 `no_graph` ablation; 04-03 attributed ≈40% of core landmark selection to graph ranking.

Action: none; plumbing.

8 Step 7 — reader grading + score fusion

Inputs: question, pool, edges, budget m.

constant	value	provenance
grading set	top-$2m$ by cosine ∪ top-$2m$ by in-degree	hand-set width
roles	6 labels (foundational . . . unrelated)	design
relevance	$-2 \cdots +2$	design
fused score	cos $\times (rel+2)/4$	**measured** (02-05 Pareto)
$rel=-2$ handling	excluded from ranking, kept as connector	**measured** (02-05)
cache	`grade_cache.db` by (qhash, arxiv_id)	determinism device
`CONCURRENCY`	4	hand-set

Algorithm. Grade only the candidate set (cost scales with m, not pool size), cache-first; rescore every graded paper by cos $\times (rel+2)/4$.
Why. The in-degree arm of the candidate set exists because ancestor-injected hubs (cosine ≈0.30–0.35) would miss a pure-cosine cut, yet are exactly the papers ranking promotes. Fusion sinks semantically-close-but-irrelevant papers.
Backed? **PARTIAL**. Fusion: **BACKED** (02-05: unrelated landmarks 5.7% → 0, recall held; LLM-as-reranker is standard practice in the literature). The $2m$ width: **UNBACKED** — and it matters more now, because trajectory magnets can only come from *graded* papers; a true reference outside the top-$2m$ cut can never become a magnet.

Action: add one battery column: fraction of core refs that entered the grading set at all. If < 1, widen $2m$ before blaming the sampler for misses.

9 Step 8 — landmark selection (trajectory hybrid over PPR)

Inputs: graded pool, edges, m, frontier spec.

constant	value	provenance
`QUOTA` q	8 consensus picks	**measured-ish** (battery Pareto at q=8)
`SEEDS`	10 fixed-seed runs	hand-set
`GAMMA`	0.4 between-lineage decay	hand-set
`MAX_LINEAGES`	50 per run	hand-set cap
thread width k	1 (linear threads)	**measured** (decay covers facets)
per-run budget	$3q$ collected before stop	hand-set
magnets	$rel \geq 2$, weight $(rel+2)/4$	design
PPR base	fills $m-q$; α=0.85, 80 iters	classic defaults
landmark budget m	30 / 60 / 80 / 120 by tier	fast m=60 **measured** (02-07)

Algorithm. Magnets = reader-graded $rel \geq 2$ papers. Each run: pick the root with the largest reachable magnet mass; descend the citation DAG with weighted-random child choice; collect magnets and continue; after each lineage, multiply collected weights by 0.4 so the next lineage is pulled toward a *different* facet. Repeat over 10 fixed seeds; the top-8 papers by collection frequency (deterministic consensus) join the PPR-ranked base ($m-8$ slots, frontier reservation from step 2). PPR order backfills if magnets run short.
Why. PPR concentrates probability on hubs; the sampler explicitly fans across lineages toward graded-relevant mass. Consensus over seeds because a single walk is high-variance and a greedy walk under-explores (both measured).
Backed? **BACKED** for the mechanism — twice: the internal selection battery (Pareto at q=8, 2026-07-14) and the external silver-qrels A/B (trajectory beats PPR 6/6 cells on core-ref landmark recall at ≥2/≥3 mentions, standard tier, capped). No literature antecedent (nearest: RL sparse reference selection *arXiv:2509.05874*, different mechanism) — this is the headline novelty. **UNBACKED** for `SEEDS`/`GAMMA`,

and q=8 was picked on the internal battery only.

Action: two items. (1) Sweep `GAMMA` $\in \{0.2, 0.4, 0.7\}$ and `QUOTA` $\in \{8, 16, 24\}$ in the battery — if trajectory keeps winning, "why not q=m?" is the first reviewer question. (2) Code the empty-magnet guard: warn + fall back to PPR when `rank=trajectory` sees no grades (the fast-tier silent-degrade trap from 04-04).

Summary — what the grand sweep should actually contain

priority	step	what	cost
1	3	`no_cluster` arm across all 20 surveys	standard (rides the battery)
2	8	`QUOTA`/`GAMMA` sweep	standard (rides the battery)
3	7	grading-set coverage column	free (logging only)
4	1	`n_pool` / `k_lexical` / `RRF_K` sweeps	cheap (deterministic, pilots)
5	1	titles-only vs title+abstract lexical arm	cheap
6	2	frontier-year eyeball per battery topic	free (logging only)
7	4	probe-saturation diagnostic	free (logging only)
—	0, 5, 6	nothing	already measured

Novelty claims riding on retrieval (bar is higher for these): per-topic frontier detection + recency quotas (step 2), hard dominant-leaf anchoring (step 3 — currently the shakiest), magnet-pulled trajectory selection (step 8 — currently the best-evidenced).